

My demo teaches rotary position embeddings through a step-by-step interactive playground. I chose RoPE because it is a central positional encoding method in modern Transformers, but it can feel abstract if students only see the formula. The main design goal was to turn the equation into visual intuition: instead of saying that RoPE rotates query and key vectors, the demo lets users drag positions and directly watch the vectors rotate.

The demo is organized as six short tabs so students can build understanding gradually. First, it shows why Transformers need position information by comparing two sentences with the same words but different meanings. Then it visualizes a 2D query/key rotation, shows how the attention dot product depends on relative position, and expands the idea to multiple frequency “clocks” across vector pairs. I included a toy attention heatmap to connect the geometry back to self-attention, while clearly labeling it as a visualization rather than real trained attention.

The final tab uses a precomputed tiny Transformer training replay on an order-sensitive task: deciding whether A appears before B. This makes the demo more purposeful because students can see that positional encoding is not just a mathematical trick; it changes what the model can learn and how well it generalizes to longer sequences. I kept the text short, used sliders/buttons instead of dense explanations, and added a fullscreen mode so the demo can be used directly in class.